

Haifa Legacy Capital (K) Series Construction Plan

0. Purpose of This Document

This report is a **blueprint for constructing monthly capital stock (K) series** for the Haifa legacy port operator (HPC) that are suitable for:

- Productivity and LP (labor productivity) analysis
- Event-study and mediation models (e.g. reform $\rightarrow$ K/L $\rightarrow$ LP)

It is written so that **future GPT chats** (or collaborators) can:

1. See **exactly** what data already exist in your project.
2. See **what additional data are needed** for improved K measurement.
3. Follow **clear, step-by-step procedures** to build several alternative monthly K paths:
4. (a) Purely linear monthly K (baseline / Version 0)
5. (b) Annual PIM + linear intra-year interpolation
6. (c) Hybrid "lumpy + smooth" monthly K (port-tailored, recommended)
7. (d) Simple "mid-year step" K

Throughout, "Haifa legacy" refers to the state-owned **Haifa Port Company** (HPC) before and around privatization.

1. Conceptual Overview: What We Are Trying to Measure

1.1 The target object: monthly K_t

Economically, we want a series K_t (monthly) that represents:

The **real, net stock of physical capital** used by the Haifa legacy port operator in month t , measured in constant prices.

This should include, in principle:

- Port infrastructure: berths, quay walls, dredging, breakwaters
- Superstructure & handling equipment: STS quay cranes, yard cranes, straddles, internal trucks
- Buildings & warehouses
- Yard surfaces and rail links
- IT systems and control towers (if included in Track B)

The **monthly K_t** will be used to construct **capital-labor ratios** K/L , especially $\ln(K/L)$, as mediators in your regression models.

1.2 Why multiple K paths?

Given data constraints (public Note-8 financials, no internal asset register), there is **no unique, assumption-free monthly K_t** . Instead, we will construct **several plausible paths** that all:

- Respect annual constraints (financial statements / PIM)
- Differ only in **intra-year timing assumptions** about when capital is installed and becomes productive

These will be used for **robustness**:

- If results are robust across paths, the empirical conclusions are less sensitive to timing assumptions.
- If they differ, the differences themselves are an informative part of your thesis.

2. Inventory of Existing Data and Code

This section lists data and scripts you already have, grouped by role.

2.1 Financial raw data (Haifa legacy)

File: `haifa_financials_raw.tsv`

- **Contents:** Annual data manually extracted from Haifa Port Company's financial statements.
- **Key economic columns:**
 - `company` — Haifa Port Company (legacy)
 - `year` — financial year (2018, 2019, 2023, 2024)
 - `ppe_net_nominal_thousands_nis_eoy`
 - Net PPE (property, plant & equipment) at year-end
 - Units: **thousands of NIS** (nominal)
 - Interpretation: **book capital stock** K_y^{book} at end of year y .
 - `depr_amort_total_thousands_nis_cf_adj`
 - Depreciation & amortization flows (cash-flow adjusted)
 - `purchase_of_fixed_assets_thousands_nis_cashflow`
 - Cash CapEx on fixed assets (gross investment candidate)
 - `purchase_of_intangible_assets_thousands_nis_cashflow`
 - CapEx on intangibles
 - `proceeds_from_realization_fixed_assets_thousands_nis`
 - Proceeds from asset disposals
- Misc. metadata: `source_pdf`, `source_pages`, `units_note` etc.

This file is a **scratchpad**: a direct mirror of Note 8 and related tables.

2.2 Clean annual financial panel (Track B, Haifa)

File: TrackB_Haifa_HPC_exc.2022.tsv

- **Contents:** Clean, structured annual panel for Haifa HPC, excluding the problematic 2022 annual (which is reconstructed elsewhere).
- **Key columns (names may vary slightly):**
 - `port` = "Haifa"
 - `company` = "Haifa Port Company Ltd."
 - `operator_or_owner` = label for legacy operator
 - `financial_year` or `year`
 - `ppe_net_nominal` — net PPE at year-end (nominal, thousands NIS)
 - `acq_fixed_assets_nominal` — gross additions to PPE
 - `disposals_nominal` — disposals of PPE
 - `depr_amort_nominal` — depreciation expense
 - Possibly: `ppe_gross_nominal`, `accum_depr_nominal`
 - Metadata: currency, units, basis/break flags

This is the **feedstock** for constructing annual capital stocks via PIM.

2.3 Current monthly K (Version 0)

File: K_B_monthly_Haifa_Legacy.tsv

- **Contents:** Monthly PPE-based K series (current pipeline output) plus splice and flags.
- **Key columns:**
 - `port`, `company`, `operator_or_owner`
 - `month` — monthly date (YYYY-MM-01)
 - `K_PPE_real_raw`
 - Monthly K obtained by **linear interpolation between annual net PPE**.
 - Currently "real" in name only: deflator is unit (1.0) unless updated.
 - `K_PPE_real_spliced`
 - Same as above, but **re-leveled (spliced)** at the 2023 privatization break so there is no artificial jump.
 - `splice_flag`
 - 0 before splice, 1 after: indicates where `K_PPE_real_spliced` differs from raw.
 - `imputed_2022`
 - 1 indicates the 2022 PPE level was reconstructed.
 - Placeholder columns (currently NaN): `K_PIM_central`, `K_PIM_low`, `K_PIM_high`, `K_services_central`.

This is your **Version 0 monthly K**: linear between annual balance-sheet points.

2.4 Mediator panel: K and (future) K/L

File: `Mediator_K_over_L.tsv`

- **Contents:** A structure that merges (or will merge) K with labor.
- **Key columns:**
 - `port`, `company`, `operator_or_owner`, `month`
 - `K_PPE_real_spliced`, `K_PPE_real_raw`, (future) `K_PIM_*`, `K_services_central`
 - `splice_flag`, `imputed_2022`
 - `hours` (currently NaN): monthly labor hours L.
 - `lnK_over_L_K_PPE_real_spliced` (currently NaN): log K/L using spliced K.
 - `lnK_over_L_K_PIM_central`, `lnK_over_L_K_PPE_real_raw`: log K/L with alternative K definitions.

This file is meant to be **the K/L mediator dataset**, once K and L are fully populated.

2.5 QA and metadata

Files:

- `qa_K_B_Haifa_Legacy.tsv`
- Contains QA metrics for the current monthly K, including splice scale factors.
- `_meta_K_B_Haifa_Legacy.json`
- Encodes the configuration used by your `Build_K` script:
 - Source file paths
 - Deflator specification
 - Splice period and scaling factor
 - Output file paths

These are crucial for understanding and modifying the K construction pipeline.

2.6 Labor and output data (used later with K)

Not all of these are in the same folder, but conceptually they exist:

- **Labor hours proxy file(s)** (e.g. `L_Proxy.tsv`)
- Port × month panel with hours `L_t` (currently a proxy) for Haifa and other ports.
- **Throughput / LP panel** (e.g. `LP_port_month_mixadjusted.tsv` or similar)
- Port × month panel with TEU, mix-adjusted LP, and other outcomes.

These will be merged with K to form the final **port × month** analysis panel.

2.7 Capital-measurement reference

File: `9789264068476-en.pdf`

- This is the **OECD Manual: Measuring Capital**.
- It describes standard PIM, productive vs wealth stocks, and user-cost based capital services.

- We will rely on this for choosing depreciation patterns and justifying methods.

3. Target K Series and Paths

We aim to construct **multiple monthly K_t series** for Haifa legacy, all rooted in the same annual PIM framework, but with different intra-year timing assumptions.

3.1 Common foundation: annual PIM capital stock, K_y^{PIM}

All monthly paths will share a **common annual backbone**:

- A PIM-based annual net capital stock K_y^{PIM} for Haifa HPC.
- Constructed from Note-8 additions, disposals, and an assumed depreciation profile.

New Output (not yet implemented): `K_B_annual_Haifa_PIM.tsv`

- Columns: `port`, `company`, `operator_or_owner`, `year`, `K_PIM_central`, `K_PIM_low`, `K_PIM_high`, plus metadata.
- Will be used as the starting point for all monthly interpolations.

3.2 Monthly K paths

We will implement at least **three** monthly paths:

1. **Path A — Purely linear (benchmark / Version 0+):**
2. Use annual K_y^{PIM} and interpolate linearly in levels (similar to current pipeline but grounded in PIM rather than raw PPE).
3. **Path B — Hybrid "lumpy + smooth" (recommended):**
4. Separate big lumpy port projects from background investment.
5. Treat big projects as step additions in the month they enter service.
6. Let background K evolve smoothly (e.g. log-linear or Denton interpolation) between annual PIM stocks.
7. **Path C — Mid-year step:**
8. Assume that all net annual investment effectively appears at mid-year.
9. Capital stock is piecewise flat within each year with a single mid-year jump.

Optionally, Path D can be defined as "PIM + purely linear" as an incremental improvement over the existing Version 0.

We will then construct **K/L and ln(K/L)** for each path and use them in regressions.

4. Step 1: Build Annual PIM Capital Stocks (K_y^{PIM})

This step replaces "net PPE directly = K" with a **proper economic capital stock** for each year.

4.1 Data required

Already available in `TrackB_Haifa_HPC_exc.2022.tsv`:

For each year y (2018, 2019, 2023, 2024):

- `ppe_net_nominal` (net PPE at year-end)
- `acq_fixed_assets_nominal` (gross additions)
- `disposals_nominal` (if available)
- `depr_amort_nominal`
- `break_or_basis_change_flag` (to track privatization/accounting changes)

Additional data to acquire / decide:

1. Deflator:

2. A price index to convert nominal additions and stocks into real terms:
 - Ideally: a **construction/port infrastructure** price index.
 - Minimum acceptable: a general capital goods or producer price index (PPI).
3. Frequency: annual (can be interpolated if you find quarterly/monthly).
4. Decision: choose a base year (e.g. 2015 or 2018) and treat that value as 100.

5. Asset lives and depreciation pattern:

6. Decide on a plausible **service life L** for aggregate port PPE:
 - E.g. $L = 30$ years for structures, 25 for equipment; or a single $L \sim 28$ years for the combined asset.
7. Decide on **depreciation pattern**:
 - Geometric (constant percentage rate): $K_y = (1 - \delta)K_{y-1} + I_y$
 - Hyperbolic or Winfrey if you want delayed retirements.
8. Choose central, low, and high service lives to generate `K_PIM_central`, `K_PIM_low`, `K_PIM_high`.

9. Initial K level (K_0):

10. Choose a starting year and anchor:
 - Option 1: set $K_{2018}^{PIM} = \text{real net PPE}_{2018}$.
 - Option 2: if you later get pre-2018 data, solve for K_0 in a quasi-steady state.
11. For now, anchoring at 2018 net PPE is acceptable and transparent.

4.2 Procedure

1. Construct real investment series I_y :

2. For each year y in Track B:

- Compute $I_{y_nominal} = acq_fixed_assets_nominal$ (optionally plus relevant intangibles).
- Divide by deflator D_y to get I_{y_real} .

3. **Compute real net PPE (wealth stocks) at year-end:**

4. $K_{y_wealth_raw} = ppe_net_nominal / D_y$.

5. These are the accounting-based net stocks.

6. **Run PIM recursion:**

7. Set initial $K_{2018}^{PIM} = K_{2018,wealth}$ or a smoothed alternative.

8. For $y \geq 2019$, compute:

- $K_y^{PIM} = (1 - \delta)K_{y-1}^{PIM} + I_y^{real}$

9. Repeat for central, low, and high δ values.

10. **Construct output:**

11. Save `K_B_annual_Haifa_PIM.tsv` with columns:

- `port`, `company`, `operator_or_owner`, `year`
- `K_PIM_central_real`, `K_PIM_low_real`, `K_PIM_high_real`
- `K_wealth_raw_real` (for comparison)
- deflator and depreciation parameters (for documentation)

4.3 What still needs to be done

- **Select a deflator** and build a small table `capital_deflator_israel.tsv` with `year` → index.
- **Choose L and depreciation pattern** and write them into a small JSON config used by the K-builder.
- **Implement a PIM script** or extend `Build_K` to:
 - Read Track B
 - Apply deflator & PIM
 - Output `K_B_annual_Haifa_PIM.tsv`

Once this is done, **all later monthly paths will be based on** `K_y^{PIM}`, not directly on raw net PPE.

5. Step 2: Build Monthly K — Path A (PIM + Linear)

This path is the closest to your current Version 0, but grounded in PIM.

5.1 Data required

- Annual PIM file: `K_B_annual_Haifa_PIM.tsv` (from Step 1).

- Existing `_meta_K_B_Haifa_Legacy.json` and `Build_K` code (to extend rather than rewrite).

5.2 Procedure

1. Merge annual PIM into monthly skeleton:

2. Create a monthly index for Haifa legacy from e.g. 2017-12 to 2024-12.

3. For each year y , mark December (or 31 Dec) with K_y^{PIM} .

4. Interpolate linearly between year-end PIM stocks:

5. For each interval $(y-1) \rightarrow y$:

- For months $k = 0, \dots, 12$ between December of $y-1$ and December of y :
- $K_t^{(A)} = K_{y-1}^{PIM} + \frac{k}{12} (K_y^{PIM} - K_{y-1}^{PIM})$

6. Apply splice (if needed):

7. If privatization or accounting breaks cause level inconsistencies, apply the existing splice logic to `K^{(A)}` instead of raw PPE.

8. Output columns:

9. In an updated `K_B_monthly_Haifa_Legacy.tsv`, add or repurpose columns:

- `K_PIM_lin_central` — monthly K from PIM+linear, central δ .
- `K_PIM_lin_low`, `K_PIM_lin_high` — if you want monthly paths for low/high lifetimes.

5.3 Status

- **To build:** need annual PIM first.
- Once annual PIM is done, this is a relatively small extension of your existing linear-interpolation pipeline.

6. Step 3: Build Monthly K — Path B (Hybrid “Lumpy + Smooth”)

(Recommended main K series)

This path is port-tailored and incorporates the lumpy nature of port investments.

6.1 Data required

Already available / partly available:

- Annual PIM file `K_B_annual_Haifa_PIM.tsv` (Step 1)
- Note-8 annual additions and, separately, your knowledge of major port projects from ARs, Port Authority docs, news.

To be constructed:

1. Project-level table for big investments

Create a file, e.g. `Haifa_big_projects.tsv`, with **one row per major discrete port project**:

Column name	Description
<code>project_id</code>	Short ID, e.g. <code>HAF_P1</code>
<code>port</code>	"Haifa"
<code>operator_or_owner</code>	Usually "Haifa-Legacy (HPC)" for your horizon
<code>description</code>	e.g. "2 new STS cranes", "Berth 3 extension"
<code>commissioning_month</code>	Date <code>YYYY-MM-01</code> when project entered service
<code>investment_year</code>	The financial year where most CapEx is recorded
<code>cost_nominal_est</code>	Estimated nominal cost (thousands NIS)
<code>cost_real_est</code>	Cost in base-year real NIS (after deflating)
<code>asset_life_years</code>	Expected life (e.g. 25 years for cranes, 35–40 for berths)
<code>source</code>	Reference (annual report page, gov document, etc.)
<code>notes</code>	Any caveats (e.g. cost approximate, multi-year construction)

You will need to **scrape ARs & port documents** to fill this in:

- Look for sections on "major investments", "capital projects", "expansion", "new cranes".
- Approximate commissioning month based on the earliest statement like "came into operation in Q3 2019".
- Approximate cost from stated amounts or proportion of Note-8 additions.

• Background investment calculation

For each year y , define:

- `I_y_total` — total note-8 additions (deflated): already used in PIM.
- `I_y_big` = `sum(cost_real_est for projects p with investment_year = y)`.
- `I_y_bg` = `I_y_total - I_y_big`.

We must ensure that `I_y_bg ≥ 0` (if not, re-allocate costs or adjust estimates).

6.2 Conceptual decomposition

We will decompose total K into:

$$K_t^{(B)} = K_{big,t} + K_{bg,t}$$

- $K_{big,t}$: capital from major discrete projects
- $K_{bg,t}$: background capital from the remainder of investment

The annual PIM logic must be consistent with this decomposition:

- The annual PIM stock K_y^{PIM} should equal the sum of big-project PIM stocks plus background PIM stocks at year-end.

6.3 Procedure: Big-project capital $K_{big,t}$

For each project p in `Haifa_big_projects.tsv`:

1. Compute a **monthly depreciation rate** $\delta_{p,month}$ from its asset life L_p and chosen pattern (usually geometric):
2. Suppose annual $\delta_{p,annual}$ is chosen such that the expected life is L_p years.
3. Then set $\delta_{p,month} = 1 - (1 - \delta_{p,annual})^{1/12}$.
4. Define monthly capital from this project, $K_{big,t}^{(p)}$:
5. For months t before commissioning month t_p :
 - $K_{big,t}^{(p)} = 0$ (under construction, not in productive stock).
6. For $t = t_p$:
 - $K_{big,t_p}^{(p)} = I_p^{real}$, where $I_p^{real} = cost_real_est$.
7. For $t > t_p$:
 - $K_{big,t}^{(p)} = (1 - \delta_{p,month}) K_{big,t-1}^{(p)}$.
8. Sum over all projects for each month:

$$K_{big,t} = \sum_p K_{big,t}^{(p)}$$

This gives a **stepwise capital stock** that jumps in the months when big projects enter service and then declines slowly with depreciation.

6.4 Procedure: Background capital $K_{bg,t}$

We want $K_{bg,t}$ to:

- Match annual background PIM stocks at year-end.
- Evolve smoothly within each year.
- Compute annual background PIM stocks

Define **annual background capital** at year-end as:

$$K_{bg,y}^{PIM} = K_y^{PIM} - K_{big,y}^{PIM}$$

where $K_{big,y}^{PIM}$ is the year-end PIM stock from big projects:

$$K_{big,y}^{PIM} = \sum_p K_{big,Dec\ of\ year\ y}^{(p)}$$

This ensures total PIM stock decomposes as:

$$K_y^{PIM} = K_{big,y}^{PIM} + K_{bg,y}^{PIM}$$

1. Interpolate $K_{bg,t}$ between annual benchmarks

For each calendar year y (between y-1 and y):

- We know $K_{bg,y-1}^{PIM}$ at end of year y-1.
- We know $K_{bg,y}^{PIM}$ at end of year y.

We must pick a monthly path $K_{bg,t}$, t in months between these endpoints, such that:

- $K_{bg,t}$ is **smooth** in some sense.
- $K_{bg,Dec\ y} = K_{bg,y}^{PIM}$.

Two recommended options:

Option 1: Log-linear interpolation (constant growth rate)

For months k = 0,...,12 between December of year y-1 and December of year y:

$$\ln K_{bg,t} = \ln K_{bg,y-1}^{PIM} + \frac{k}{12} (\ln K_{bg,y}^{PIM} - \ln K_{bg,y-1}^{PIM})$$

This means background capital grows at a constant **percentage** rate within the year.

Option 2: Denton disaggregation (advanced)

Use a Denton method (or similar) to allocate annual background capital across months so that:

- Annual averages or year-end values match $K_{bg,y}^{PIM}$, and
- Month-to-month changes in the growth rate of $K_{bg,t}$ are minimized.

This requires more coding but yields a theoretically clean and very smooth series.

1. Combine to get hybrid K

For each month t:

$$K_t^{(B)} = K_{big,t} + K_{bg,t}$$

In code, this would be stored as a new column in `K_B_monthly_Haifa_Legacy.tsv`, e.g.:

- `K_PIM_hybrid_central`

6.5 Status and tasks

Data to acquire:

- Populate `Haifa_big_projects.tsv` by reading ARs, port authority docs, etc.
- Decide on cost estimates and commissioning months.
- Decide on asset lives by project type.

Code to write or adapt:

- A script to:
- Read `K_B_annual_Haifa_PIM.tsv` and `Haifa_big_projects.tsv`.
- Compute $K_{big,t}^{(p)}$ and $K_{big,t}$.
- Compute $K_{bg,y}^{PIM}$ and interpolate $K_{bg,t}$.
- Form $K_t^{(B)}$ and save as `K_PIM_hybrid_central` in `K_B_monthly_Haifa_Legacy.tsv`.

This path is your **preferred K for empirical work** once implemented.

7. Step 4: Build Monthly K — Path C (Mid-Year Step)

This is a simpler alternative path that embodies "lumpy" timing without project-level data.

7.1 Data required

- `K_B_annual_Haifa_PIM.tsv` (Step 1).

7.2 Procedure

For each year y , we define that **all net investment flows are recognized at mid-year**.

A simple implementation:

1. For months Jan–Jun of year y :
2. Set $K_t^{(C)} = K_{y-1}^{PIM}$.
3. For months Jul–Dec of year y :
4. Set $K_t^{(C)} = K_y^{PIM}$.

This yields a piecewise-constant K that jumps once per year.

A slightly more accurate version can adjust for half-year depreciation, but the key idea is the same: **lumpy mid-year capital formation**.

7.3 Output

- New column in `K_B_monthly_Haifa_Legacy.tsv`:
- `K_PIM_midyear_central`.

7.4 Status

- Requires only annual PIM; easy to implement once Step 1 is done.

8. Step 5: Merge Monthly K with Labor to Form K/L

Once K paths are built, we add labor and compute K/L.

8.1 Data required

- `K_B_monthly_Haifa_Legacy.tsv` with:
- `K_PPE_real_spliced` (current PPE-based K)
- `K_PIM_lin_central`, `K_PIM_hybrid_central`, `K_PIM_midyear_central` (new paths)
- Labor file (e.g. `L_Proxy.tsv`):
- Port = "Haifa"
- Monthly `hours` for Haifa legacy.

8.2 Procedure

1. Merge K and L

2. Join `K_B_monthly_Haifa_Legacy.tsv` with your labor panel on:

3. `port = "Haifa"`

4. `month`

5. (and, if needed, `operator_or_owner` to select the legacy period)

6. Compute K/L for each path

Add columns in `Mediator_K_over_L.tsv` such as:

- `lnK_over_L_K_PPE_real_spliced`
- $= \ln(K_{PPE_real_spliced}) - \ln(hours)$
- `lnK_over_L_K_PIM_lin_central`
- `lnK_over_L_K_PIM_hybrid_central`
- `lnK_over_L_K_PIM_midyear_central`

You may choose to keep all in a single mediator file or create multiple mediator files by path.

1. Flag scenarios

Add or update flags:

- `k_path` (string): `"PPE_spliced"`, `"PIM_lin"`, `"PIM_hybrid"`, `"PIM_midyear"`.
- Alternatively, keep one wide file with many columns and use column names directly in analysis.

8.3 Output

- Updated `Mediator_K_over_L.tsv` with fully populated $\ln(K/L)$ mediators for multiple K paths.

9. Step 6: Using K Paths in Regressions and Sensitivity Analysis

Once $\ln(K/L)$ is available for multiple K paths, you will:

9.1 Main LP regressions

For each path $v \in \{\text{PPE_spliced}, \text{PIM_lin}, \text{PIM_hybrid}, \text{PIM_midyear}\}$:

- Estimate:

$$\ln LP_{it} = \alpha_i + \gamma_t + \beta^{(v)} \ln(K/L)_{it}^{(v)} + X_{it}\theta + \varepsilon_{it}$$

- Compare $\beta^{(v)}$ across v.

9.2 Event study and mediation

- Stage 1: effect of competition / privatization on $\ln(K/L)^v$

$$\ln(K/L)_{it}^{(v)} = f(\text{event time}) + \text{controls} + \mu_i + \tau_t + u_{it}$$

- Stage 2: effect of $\ln(K/L)^v$ on $\ln(LP)$, controlling for events.

Compare across v to see whether your conclusions about **capital deepening as a channel** are sensitive to the assumed K path.

9.3 Reporting in the thesis

You will:

- Present $\beta^{(v)}$ and mediated effects for at least three K paths.
- Show that results are **qualitatively similar** (or highlight differences and interpret them).
- Explicitly note that K is a **constructed object** and that you have checked timing-assumption robustness.

This turns your current discomfort into a **methodological contribution**: you didn't just pick one arbitrary interpolation, you systematically examined realistic alternatives.

10. To-Do Checklist by Step

This final section is a practical checklist of what still needs to be acquired or coded.

10.1 Data to acquire

1. **Capital deflator series** for Israel:
2. At least annual index for relevant capital goods.
3. Source: CBS Israel, OECD, or other.
4. **Big project details for Haifa legacy**:
5. Read ARs (2018–2024), MoT and Port Authority docs, news.
6. Build `Haifa_big_projects.tsv` with:
 - Commissioning month
 - Estimated cost
 - Asset life
7. **Confirm or refine labor hours series**:
8. Ensure `L_Proxy.tsv` (or equivalent) is ready for Haifa legacy and aligned with your month index and operator labels.

10.2 Code to write or adapt

1. **PIM builder (annual)**:
2. Input: `TrackB_Haifa_HPC_exc.2022.tsv`, deflator, depreciation config.
3. Output: `K_B_annual_Haifa_PIM.tsv`.
4. **Monthly K paths**:
5. Extend/modify `Build_K` so that for Haifa legacy it can:
 - Path A: PIM + linear (`K_PIM_lin_central`).
 - Path B: PIM + lumpy+smooth using `Haifa_big_projects.tsv` (`K_PIM_hybrid_central`).
 - Path C: PIM + mid-year step (`K_PIM_midyear_central`).
6. **Mediator builder**:

7. Script to merge K and labor, compute $\ln(K/L)$ for each K path, and save into `Mediator_K_over_L.tsv`.

10.3 Documentation to add

- Write a short markdown/LaTeX section for your thesis methods chapter summarizing:
 - The PIM setup (deflator, lives, depreciation).
 - The K path definitions.
 - The robustness strategy.
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11. How to Use This Document in Future GPT Chats

When you open a new GPT chat to, for example, extract big projects or implement PIM, you can:

1. Upload this report along with the relevant data files.
2. Say something like:

“This document (`Haifa Legacy K-Series Construction Plan`) describes my K construction strategy. We’re currently at Step X. Please implement the procedure described in section Y, using these files: ...”

1. Ask GPT to:
2. Generate code (Python, R, Stata) that follows the specific steps.
3. Help populate `Haifa_big_projects.tsv` by reading ARs and inferring project details.
4. Produce draft text for your thesis methods section based on Sections 4–9.

This way, every new chat has a full specification of **what K is supposed to be** and **where you are in the pipeline**, ensuring consistency across your work.